

Multi Metric Analysis of CMOS and Adiabatic Logic Digital Circuits

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Abstract — Rapid evolution of digital logic circuits, low-power, and high-performance design methodologies are essential. Although CMOS technology is popular, its application is becoming challenging, especially with technology scaling. The application of adiabatic logic circuits, which use energy recovery techniques, is an emerging trend in reducing switching power losses in digital logic circuits. In this paper, a comparative analysis of conventional CMOS logic circuits with adiabatic Positive Feedback Adiabatic Logic (PFAL) logic circuits using various design metrics is presented. Combinational and sequential logic circuits, including an inverter, multiplexer, demultiplexer, adder, encoder, decoder, flip-flops, are designed using Verilog HDL, and system-level synthesis and evaluation are performed using Xilinx Vivado tools to obtain power consumption, FPGA resource utilization, and timing delay of the logic circuits. At the same time, transistor-level simulation of basic logic gates of CMOS and PFAL logic families using NGSPICE tools is performed to obtain the logical functionality of the adiabatic logic gates, as well as the supply current, thereby estimating the energy consumption of the logic circuits. The obtained design metrics are organized using an interactive dashboard to compare the design methodologies clearly. The results show that adiabatic logic circuits consume much less power than conventional CMOS logic circuits, with some penalty in terms of increased hardware complexity, thereby validating the power-area trade-off of the adiabatic logic circuits. The framework of validating the design using both circuit-level simulation tools and system-level synthesis tools is an emerging trend in evaluating the design of low-power digital logic circuits.

Keywords- CMOS logic, adiabatic logic, PFAL, Low-power digital circuits, NGSPICE, Xilinx Vivado, Power and area trade-off.

I. INTRODUCTION

The rising demand for power-efficient and high-performance digital devices has further emphasized the significance of low-power design methodologies. With the scaling of technology, the problem of power dissipation has become one of the major challenges for modern integrated circuit design. Although Complementary Metal-Oxide-Semiconductor (CMOS) logic is the most widely used logic family due to its reliability and scalability, it mainly consumes dynamic power due to switching activity, especially for high-speed and large-scale digital devices.

To overcome the problem of power dissipation due to switching activity, adiabatic logic has been introduced as an alternative logic design approach. Adiabatic logic aims at reducing the power dissipation due to switching activity using the concept of energy recovery. In adiabatic logic, unlike conventional CMOS logic that consumes power during the charge and discharge of the nodes, a time-varying power clock is used for the operation of the logic circuit, thus recovering a portion of the stored charge.

Positive Feedback Adiabatic Logic (PFAL) has become one of the prominent adiabatic logic families due to its improved performance in terms of energy efficiency and reliability.

In this work, a complete multi-metric comparative analysis of the CMOS logic family and the adiabatic logic family of digital logic circuits is provided. Various combinational and sequential logic circuits such as the inverter, multiplexer, demultiplexer, adder, encoder, decoder, flip-flops are designed using the Verilog HDL language. The performance of the logic circuits is evaluated at the system level as well as the circuit level. The circuit level implementation of the logic circuits and the evaluation of the metrics such as power dissipation, FPGA resources used, and delay are performed using the Xilinx Vivado environment.

The main aim of the article is to evaluate the power-area-performance trade-offs of the CMOS logic family of digital logic circuits and the adiabatic logic family of digital logic circuits.

The residual section of the paper comprise as, section II holds ease of use, section III encloses the literature survey, section IV illustrate the methodology of the projected work, section V involves the experimental results, section VI comprises the conclusion, finally acknowledgment is incorporated in section VII.

II. EASE OF USE

- A. Reduced implementation complexity through the use of Behavioral Verilog for circuit modeling, enabling straightforward design and modification of digital circuits without extensive structural redesign.
- B. Circuit performance validation using NGSPICE ensures that transistor-level simulation is carried out under controlled conditions, allowing for reliable extraction of parameters such as power and delay.
- C. Export of simulation results in CSV format allows for simple handling of the data and integration with the Python-based framework for analysis.
- D. The modular structure enables clear visualization and comparison of performance metrics, and can be easily extended to analyze additional circuits.

III. RELATED WORKS

The concept of adiabatic switching in digital systems was introduced by Athas et al. [1]. The authors showed that it is possible to reduce the dissipation of energy during charging and discharging of circuit nodes using reversible approaches. This is important because it laid the foundation for the theoretical and practical aspects of low-power adiabatic circuit design.

A comparative analysis of CMOS logic and different adiabatic logic circuits for an 8:1 multiplexer was performed by Singh et al. [2]. The research proved that using adiabatic logic, it is possible to reduce power consumption, thus making it suitable for modern digital systems.

Kumar and Singh [3] discussed the performance evaluation of PFAL logic circuits. The research proved that PFAL logic is more suitable for modern digital systems because it is more energy-efficient than traditional CMOS logic, especially during switching transitions. Nabi and Islam [4] further explored power-efficient adiabatic circuit designs and showed measurable reductions in power dissipation in digital circuits.

Bhunia et al. [5] carried out a comparative study of CMOS and adiabatic logic for MUX and DEMUX

circuits. Their results indicated lower power consumption in adiabatic logic, but also pointed out increased circuit complexity as a key limitation. Similarly, Kumar and Kumar [6] analyzed various challenges associated with adiabatic logic, including design complexity, clocking requirements, and practical implementation issues.

Malarkodi proposed optimization techniques for energy-efficient VLSI circuits for next-generation IoT devices [7]. This paper emphasized the significance of low-power design techniques, adiabatic logic, in improving energy efficiency in modern electronic systems.

Dokic presented a comprehensive review on energy-efficient CMOS design techniques, with a focus on power reduction techniques for dynamic and leakage power reduction in CMOS circuits [8]. This paper reveals that though CMOS optimization can be used for minimizing power consumption, still it has limitations in achieving efficient results compared with energy-recovery-based logic styles. Chattopadhyay et al.[9] Proposed energy-efficient adiabatic logic circuits for portable electronic devices and demonstrated the suitability of adiabatic logic circuits for portable devices due to efficient power consumption.

Vanlalchaka et al. [10] Proposed a recent adiabatic logic technique with the aim of improving energy efficiency in digital circuits. This paper reveals that modified adiabatic logic can be used for minimizing energy consumption in digital circuits, thus showing the advancements in adiabatic logic techniques.

IV. METHODOLOGY

The comparison between CMOS and adiabatic logic circuits in this work is carried out using multiple performance parameters. At the beginning, the selected digital circuits were written in Behavioral Verilog to check whether the logic works correctly and to keep the design flexible. Once the functionality was verified, the same circuits were taken to the transistor level so that their electrical behaviour could be examined more closely. For a fair comparison, both CMOS and adiabatic versions were designed under the same operating conditions. NGSPICE was used to run the simulations and observe circuit behavior at the device level. The same input patterns and supply values were applied in both cases to avoid variations in the results. After the simulation, the key parameters were observed. The parameters included power consumption, propagation delay, and switching characteristics. The observed values were stored in a CSV file. This can be easily analyzed by making use of Python code. This would be helpful to compare the observed values in an organized manner. In this case, instead of focusing on a single parameter, power consumption and delay will be considered. This would give a better idea about how power consumption and performance trade off in CMOS and adiabatic logic circuits

A. Block Diagram

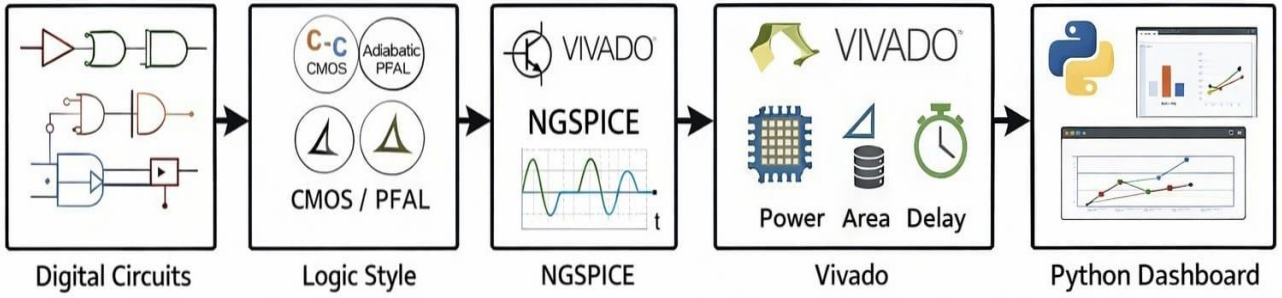


Fig. 1. Block Diagram of Simulation flow and output Visualization

Fig. 1 shows the block diagram used for multi-metric analysis of digital circuits. The process is divided into stages such as digital circuit design, logic style selection, and simulation using NGSPICE, further processing of results.

a) *Digital Circuits*: This block represents the design of fundamental digital circuits such as logic gates, combinational circuits and sequential circuits. These circuits are described at the behavioral modelling in verilog, to ensure design flexibility and functional verification. These circuits serve as the base structures for performance evaluation under different logic styles. The correctness of logical operations is verified before proceeding to further stages of implementation and analysis.

b) *Logic Style*: The logic style block defines the implementation approach of the designed circuits. In this work, circuits are implemented using conventional CMOS logic and adiabatic logic techniques such as Energy Recovery Clocked Logic (ECRL) and Positive Feedback Adiabatic Logic (PFAL). This stage determines how switching behavior and power dissipation characteristics differ between traditional and energy-recovery logic families.

c) *NGSPICE*: This block performs transistor-level simulation and validation of the implemented circuits. It is used to analyze electrical characteristics under uniform operating conditions. Through waveform analysis, key parameters such as average power consumption, propagation delay, and switching behavior are extracted. This stage ensures accurate and reliable performance validation of both CMOS and adiabatic implementations.

d) *Vivado*: This phase utilizes Vivado tool for the synthesis of the design and observation of the design's behavior at the implementation level. This phase is useful for the acquisition of data such as power consumption, area utilization, and delay. The synthesized design will facilitate easy comparison between CMOS and adiabatic logic from a hardware perspective.

e) *Python Dashboard*: The final phase involves the processing of the results utilizing a Python-based dashboard for easier analysis and understanding of the

results. The simulation and synthesis results will be stored in CSV format, which will be used for easier organization and comparison. The results will be displayed graphically for easier understanding of the difference between CMOS and adiabatic logic. This method can also be used for the inclusion of additional circuit designs.

B. Flow Chart:

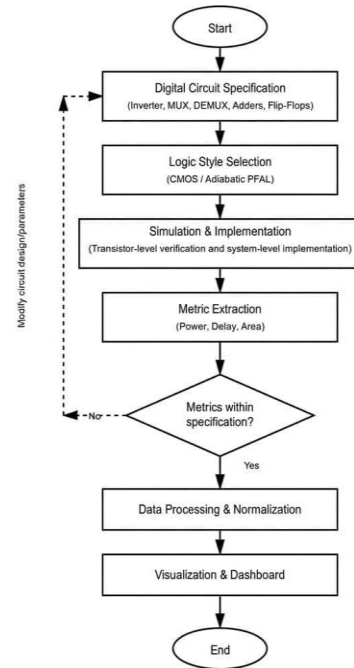


Fig. 2. Flowchart of Proposed Multi Metric Evaluation

The above Fig. 2 flowchart indicates a step-by-step process in the analysis of digital circuits using the multi-metric evaluation method. First, a logic style is chosen for implementing the design. This could be either adiabatic logic or CMOS logic. After this step, the design is developed and simulated to observe how it behaves under different input conditions. From this simulation, the performance is evaluated in terms of power consumption,

delay, and area. These results are then compared with the required performance standards. If those results are not satisfactory, necessary changes are made to the digital circuits and the simulation is repeated. This process continues until the performance meets the expected criteria. Once acceptable results are achieved, the data is organized to maintain consistency in evaluation. Finally, the results are shown through a dashboard for easier interpretation, and the process concludes.

C. Implementation:

Digital circuits were implemented using both CMOS and adiabatic logic styles to study their practical behavior. At the initial stage, the circuits were written in Behavioral modelling to check whether the logic functions correctly. Combinational and sequential modules were developed in a modular form so that changes could be made easily in Verilog. For CMOS, transistor level implementations were created using Vivado, and the corresponding netlists were simulated in NGSPICE to observe electrical characteristics under defined conditions. Care was taken while applying input patterns so that similar switching activity was maintained across all designs. In the case of adiabatic logic, the circuit structure was modified to include energy recovery during switching. Proper attention was given to clocking and signal transitions to ensure correct operation during simulation. The results obtained from these simulations were stored in CSV format and processed in Python using Visual Studio Code. Libraries such as pandas and matplotlib were used to organize the data and generate graphical comparisons, which were then presented through a dashboard for better interpretation of results and ensured that the extracted performance values were reliable.

V. RESULTS

Divide it into two subsections:

A. Vivado Results

B. Dashboard-Based Comparative Analysis:

A. Vivado Results:

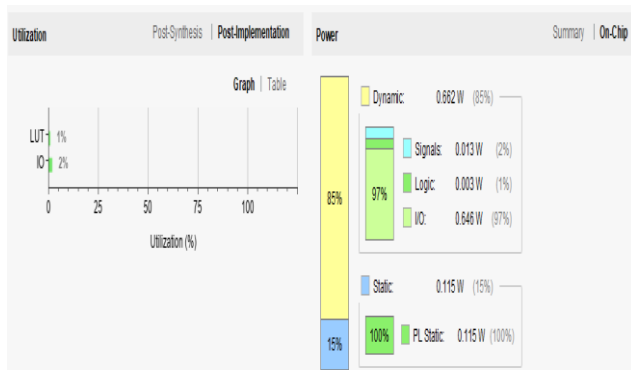


Fig. 3. Vivado synthesis results showing utilization and Power report of the implemented cmos half adder.

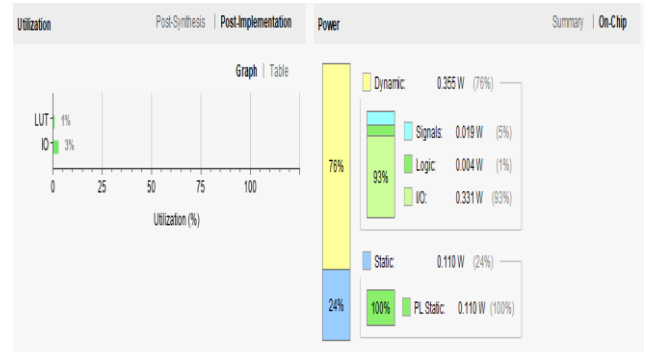


Fig. 4. Vivado synthesis results showing utilization and Power report of the implemented adiabatic half adder.

Figs. 3 and 4 illustrate the Vivado post-implementation synthesis results of the CMOS and adiabatic half adder circuits, respectively. The reports present the resource utilization and on-chip power distribution, highlighting the differences in dynamic and static power characteristics, between the two logic styles.

Vivado was used to obtain power, area, and delay report for all implemented circuits. Due to space limitations, the report for the Half Adder (CMOS and Adiabatic) is presented as a sample, while the complete results are summarized through the dashboard-based comparative analysis

B. Dashboard-Based Comparative Analysis:

Power Comparison (CMOS vs Adiabatic PFAL)

Dynamic power consumption measured in Watts (W)

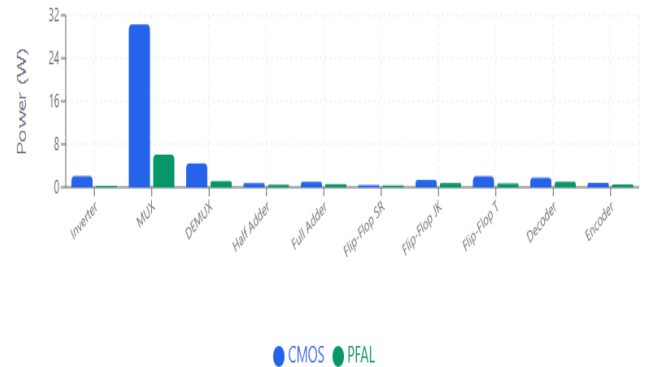


Fig. 5. Power comparison of CMOS vs. Adiabatic logic

The above Fig. 5 shows the dynamic power consumption comparison between CMOS and Adiabatic logic styles across multiple digital circuits.

Area Utilization Comparison

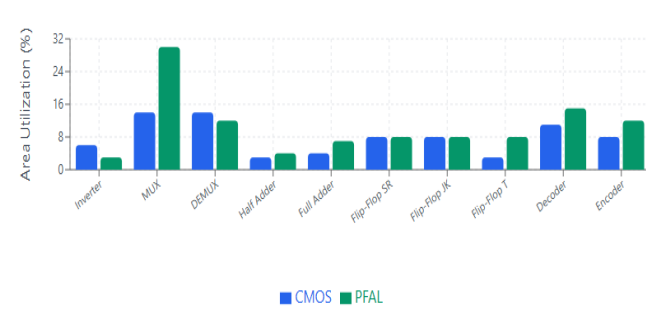


Fig. 6. Area utilization of CMOS vs. Adiabatic logic

The above Fig. 6 shows the area utilization of CMOS and Adiabatic logic implementations for different digital circuits. The graph indicates that while adiabatic reduces power consumption, it may introduce variations in area usage depending on the complexity of the logic block.

Logic Block Comparison Table

Logic Block	CMOS Power (W)	PFAL Power (W)	CMOS Area (%)	PFAL Area (%)
Inverter	2.096	0.245	6.0	3.0
MUX	30.308	6.081	14.0	30.0
DEMUX	4.446	1.159	14.0	12.0
Half Adder	0.777	0.465	3.0	4.0
Full Adder	1.019	0.595	4.0	7.0
Flip-Flop SR	0.381	0.311	8.0	8.0
Flip-Flop JK	1.371	0.803	8.0	8.0
Flip-Flop T	2.076	0.718	3.0	8.0
Decoder	1.818	1.023	11.0	15.0
Encoder	0.827	0.541	8.0	12.0

Fig. 7. Comparison table of power and area

The above Fig. 7 presents the comparative analysis of power consumption and area utilization for various digital logic blocks implemented using CMOS and Adiabatic logic styles. The table highlights the significant reduction in power achieved by PFAL implementations, along with variations in area utilization across different circuit modules.

VI. CONCLUSION

A multi-metric comparison between CMOS and Adiabatic logic digital circuits was carried out using both circuit-level and system-level analysis. NGSPICE was used to study the circuits at the transistor level, allowing observation of electrical behavior and current characteristics. In contrast, Vivado was used at the design level to obtain parameters such as power, area, and timing for FPGA-based implementations. The results indicate that PFAL logic can reduce switching energy compared to conventional CMOS, although it introduces a moderate increase in resource usage. The data obtained from both tools was organized in CSV file format and executed in Python by using libraries, where the results were compared and visualized through a dashboard for different circuits. This analysis indicates that adiabatic logic is a viable option for low-power digital circuits, while clearly highlighting the trade-offs between power, area, and performance.

VII. ACKNOWLEDGMENT

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REFERENCES

- [1] Athas, W. C., Svensson, L. J., Koller, J. G., Tzartzanis, N., & Chou, E. Y. (1994). Low-power digital systems based on adiabatic-switching principles. *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, 2(4), 398–407.
- [2] Singh, Shivani, et al. "Design and Comparative Analysis of CMOS and Various Adiabatic Logic Circuits of 8: 1 Multiplexer." NIELIT's International Conference on Communication, Electronics and Digital Technologies. Singapore: Springer Nature Singapore, 2025.
- [3].Kumar, R., & Singh, J. (2012). Performance analysis of PFAL adiabatic logic circuits. *International Journal of Computer Applications*, 47(14), 5–10.
- [4]. Nabi, A., & Islam, S. (2016). Power-efficient adiabatic logic design for low-power VLSI applications. *International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering*, 5(5), 4023–4030.
- [5].Bhunia, S., Debnath, P., & Ghosh, S. (2018). A comparative study on CMOS and adiabatic logic implementations for MUX and DEMUX circuits.

[6].Kumar, Dinesh, and Manoj Kumar. "Comparative analysis of adiabatic logic challenges for low power CMOS circuit designs." *Microprocessors and Microsystems* 60 (2018): 107-121.

[7]. Malarkodi, K. "Energy-Efficient VLSI Circuit Optimization for Next-Generation IoT Devices." 2025 8th International Conference on Trends in Electronics and Informatics (ICOEI). IEEE, 2025.

[8]. Dokic, B. L. "A review on energy efficient CMOS digital logic." *Engineering, Technology & Applied Science Research* 3.6 (2013): 552-561.

[9]. Chattopadhyay, A., Das, R., & De, D. (2011). Design and analysis of energy-efficient adiabatic logic circuits for portable electronics. *International Journal of Electronics*, 98(10), 1371–1385.

[10] Vanlalchaka, Reginald H., et al. "A novel adiabatic logic technique for low-power circuit applications." *Microsystem Technologies* 31.2 (2025): 611-630.